Lab 4. Broken PR fixer, unattended

A failing branch in. A green one out, or an honest explanation of why not.

18 minutes, not 25. Energy is lowest. Keep moving.

Work from `labs/lab4_fixer`.

What you will build

Two functions in `loop.py`:

```
def summarize_failure(run_result: RunResult) -> str: ...
def repair_until_green(contract: Contract, budget: int = 3) -> dict: ...
```

Already shipped: `contract.py` (the repo contract, 344 lines).

Final outcome. A loop that can give up out loud.

Why this lab exists

Nobody is watching. Exits matter more than successes.

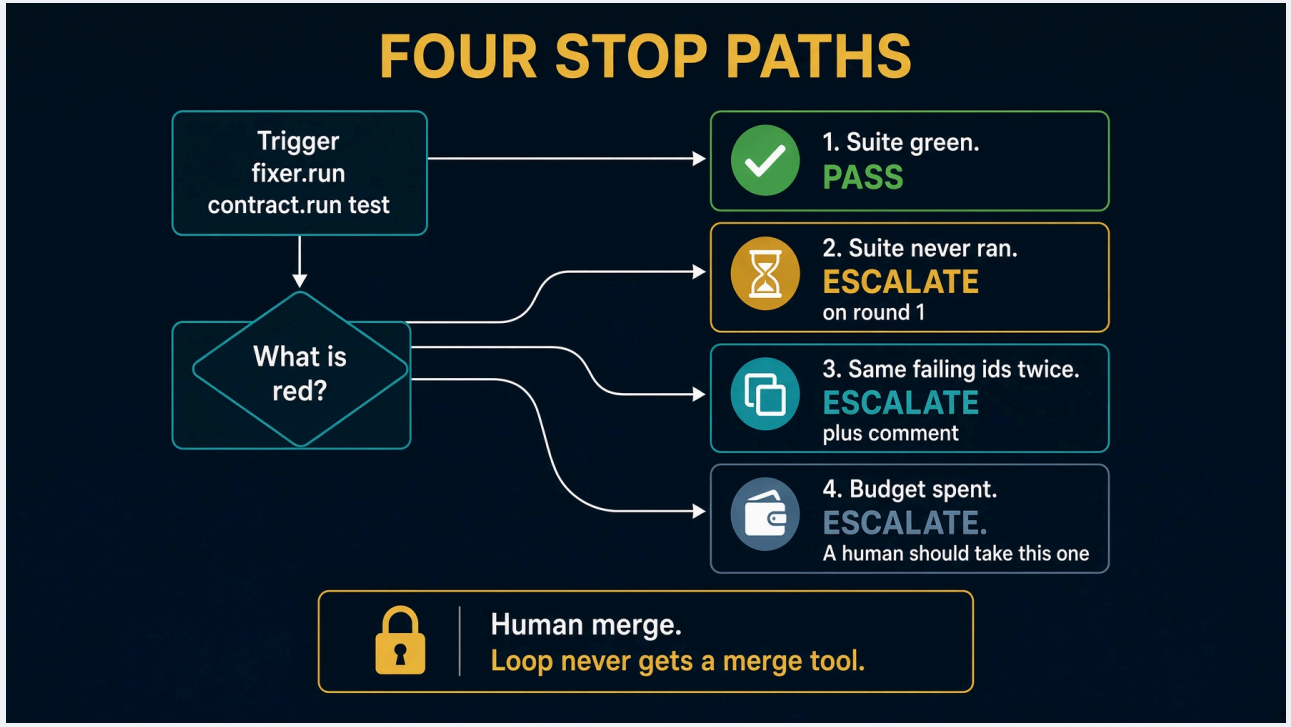
Giving up is allowed. Giving up silently is the bug.

A model that may both act and verify can invent its own evidence. That is why the receipt exists.

Learning objectives

- Implement `summarize_failure` so the error is not lost in the middle of a log
- Implement `repair_until_green` with four stop paths
- Stash Module 2 work **before** `broken-pr` checkout
- Validate `--doer none` leaves "A human should take this one"
- Troubleshoot a loop that wants to delete an earlier lab's work

Starting architecture



Cast is three roles, not five. No planner. The work is defined by what is red.

Stash first. Say it out loud

```
git -C ../../work/northwind-field-crm stash --include-untracked
```

`checkout()` on a dirty tree refuses rather than deleting Lab 2 work:

```
cannot switch to broken-pr: northwind-field-crm still holds
the work from an earlier lab.
  keep it:     git -C ... stash --include-untracked
  discard it:  git -C ... checkout -- . && git -C ... clean -fd
```

That refusal is on brand.

CRM branches

BRANCH	STATE
`main`	no due dates, ~75% coverage, below floor
`known-good`	due dates implemented, everything green
`broken-pr`	dropped the null guard, one test red

`--branch broken-pr` is what makes this real. Point it at a green branch and it reports a pass and proves nothing. Same shape as the red gate.

The stub

```
def summarize_failure(run_result: RunResult) -> str:
    raise NotImplementedError("fill me in")

def repair_until_green(contract: Contract, budget: int = 3) -> dict:
    raise NotImplementedError("fill me in")
```

Fill only `loop.py`.

`summarize_failure`

Sending the whole log puts the failure in the middle. Lost in the Middle, third time today.

```
def summarize_failure(run_result) -> str:
    failed = sorted(run_result.junit.failed_ids)
    lines = [f"{len(failed)} failing: {', '.join(failed[:5])}"] if failed else ["the suite is red"]
    error = ERROR_IN_OUTPUT.search(run_result.output or "")
    if error:
        lines.append(error.group(0).strip()[:200])
    return "\n".join(lines)
```

Four stop paths

1. Suite green → pass
2. Suite never ran → escalate on round 1
3. Same failing ids twice → escalate, leave a comment
4. Budget spent → escalate, "A human should take this one"

Scope violations escalate even if the suite is green. Reaching green by editing the failing test is not a fix.

Commands. Live first.

Saturday: `task test` imports `loop`.

Stash Lab 2 work first. Then demo the live fixer from the Agent SDK port:

```
git -C ../../work/northwind-field-crm stash --include-untracked
cd ../../solutions/sol4_fixer_agent_sdk
task setup
task test
task table          # judge writes must print no
task clone
task reset          # checkout broken-pr. Refuses a dirty clone.
task run -- --branch broken-pr --doer reference
task run -- --branch broken-pr --doer sdk
```

`--doer reference` needs no key. `--doer sdk` needs `ANTHROPIC_API_KEY`. `permission_mode` is `dontAsk`. Merge is never a tool.

Read `HOW_TO_RUN.md` and `E2E_PLAN.md` in that folder. The Deep Agents twin is the graph, not the repair loop.

Expected `--doer none`

```
attempt 1: 1 failing -> retry
  1 failing: tests.test_overdue::test_overdue_ignores_tasks_with_no_due_date
attempt 2: 1 failing -> escalate

gate: escalate
The fixer gave up.
A human should take this one.
```

`--doer reference` against `broken-pr`: `gate: pass`, files copied from `known-good` inside `app/**` only.

MAST, in one slide

Most agent failures are not model failures. 1,642 traces. Cemri et al. arXiv:2503.13657.

CATEGORY	SHARE	THIS HOUR
System design	41.8%	graph, scope, budget, trigger
Inter-agent	36.9%	summaries, not dumps
Verification	21.3%	pytest, receipt

Every one of those three is something you build, not something you buy.

Troubleshooting

SYMPTOM	CAUSE	FIX
Checkout refuses	Lab 2 files dirty	stash, out loud
`--doer none` is green	loop is lying	same ids twice escalate
Edited `tests/**`	scope not enforced	deny list on the coder
Silent give-up	missing comment	"A human should take this one."

Recap

Takeaways

1. Stash first.
2. Giving up is allowed. Giving up silently is the bug.
3. Merge is never a tool.
4. If you cannot read the last score, you cannot debug at 2 a.m.

The loop is the product. The prompt is not.